PROJECT PROGRESS REPORT (SESSION 2026-27)

STOCK MARKET ANALYSIS AND PRICE PREDICTION SYSTEM

Bachelor of Computer Applications (BCA)
School of Computer Applications and Technology
Galgotias University, Greater Noida, Uttar Pradesh, India

Field	Project Review Details
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Supervised By	[Enter Guide Name], Assistant Professor / Associate Professor
Reporting Period	Mid-Term Review / Progress Milestone (October, 2026)
Completion Status	85% Completed – Core Development & ML Verification Complete; Final Refinements Underway

SECTION 1: ABSTRACT & PROJECT OVERVIEW

1.1 Problem Statement

Equity price movements represent non-linear, stochastic time series governed by macroeconomic factors, corporate financial results, and market psychology. Retail investors and researchers struggle to manually parse raw numerical price feeds while simultaneously tracking breaking corporate news. Existing statistical tools fail under regime shifts, and commercial financial terminals carry prohibitive licensing fees.

1.2 Project Objectives

1. Ingest historical OHLCV data for global equities (US & Indian NSE) using free APIs without mandatory keys.
2. Compute mathematical technical indicators (SMA 20/50/200, EMA 20, RSI-14, MACD, Bollinger Bands) using pure Pandas/NumPy.
3. Architect a look-ahead-free ML pipeline with 14 engineered time-series features and strict chronological 80/20 train/test splitting.
4. Calculate quantitative evaluation metrics (RMSE, MAE, MAPE %) and forecast multi-day future

trajectories with 95% confidence intervals.
5. Implement real-time VADER NLP news sentiment scoring categorized into Bullish, Neutral, and Bearish market stances.
6. Provide an intuitive 4-tab Streamlit dashboard with interactive Plotly candlestick charting and CSV data export.

1.3 Current Progress Summary

Currently, 85% of project milestones are completed and validated. The core ingestion module, technical indicators, ML regression models, VADER sentiment engine, and Streamlit user interface are 100% operational. The remaining 15% focuses on final viva slide preparation, multi-sector asset comparisons, and thesis binding.

SECTION 2: SYSTEM REQUIREMENTS & APIS USED

2.1 Hardware & Software Stack

Category	Tool / Library	Specification	Project Function
Hardware	Processor (CPU)	Intel Core i3 / Ryzen 3+	Vectorized computations & tree training
Hardware	RAM & Network	4 GB+ RAM, Active Wi-Fi	In-memory caching & external API polling
Software	Python Runtime	Python 3.11.x (Astral)	Core execution and data science stack
Software	Web Framework	Streamlit 1.35+	Interactive UI dashboard & widget state
Software	Visualization	Plotly 5.22+	Interactive candlestick and forecast curves
Software	Data Science	Pandas 2.2+, NumPy 1.26+	DataFrames & vectorized indicator math
Software	Machine Learning	Scikit-Learn 1.4+	Random Forest, Gradient Boosting, Linear Reg
Software	Sentiment NLP	vaderSentiment 3.3.2	Valence-aware headline sentiment scoring
Software	Data Ingestion	yfinance 0.2.40+ / requests	Zero-key historical quotes & news search

Table 1: Hardware & Software Stack Specification

2.2 External Free APIs & Data Sources

API / Source	Protocol	Data Acquired	Failover & Error Handling
Yahoo Finance (yfinance)	HTTPS / Python Wrapper	Daily historical OHLCV bars & metadata	100% Free / Displays UI warning on invalid ticker

MultiIndex Column Cleaner	In-Memory Sanitizer	Normalized single-level columns	Flattens tuple headers to avoid shape/index error
Finnhub Stock API	HTTPS / REST JSON	Live corporate press releases & news	Optional token / Transparent fallback to Yahoo news
Yahoo Search News	HTTPS / JSON REST	Recent financial journalism articles	Default news feed when Finnhub key is absent
Google News RSS	HTTPS / XML RSS	Real-time ticker news headlines	Final fallback ensuring sentiment is always active

Table 2: External Free APIs & Data Sources Specification

SECTION 3: SYSTEM ARCHITECTURE & METHODOLOGY

3.1 Modular Architecture Flowchart

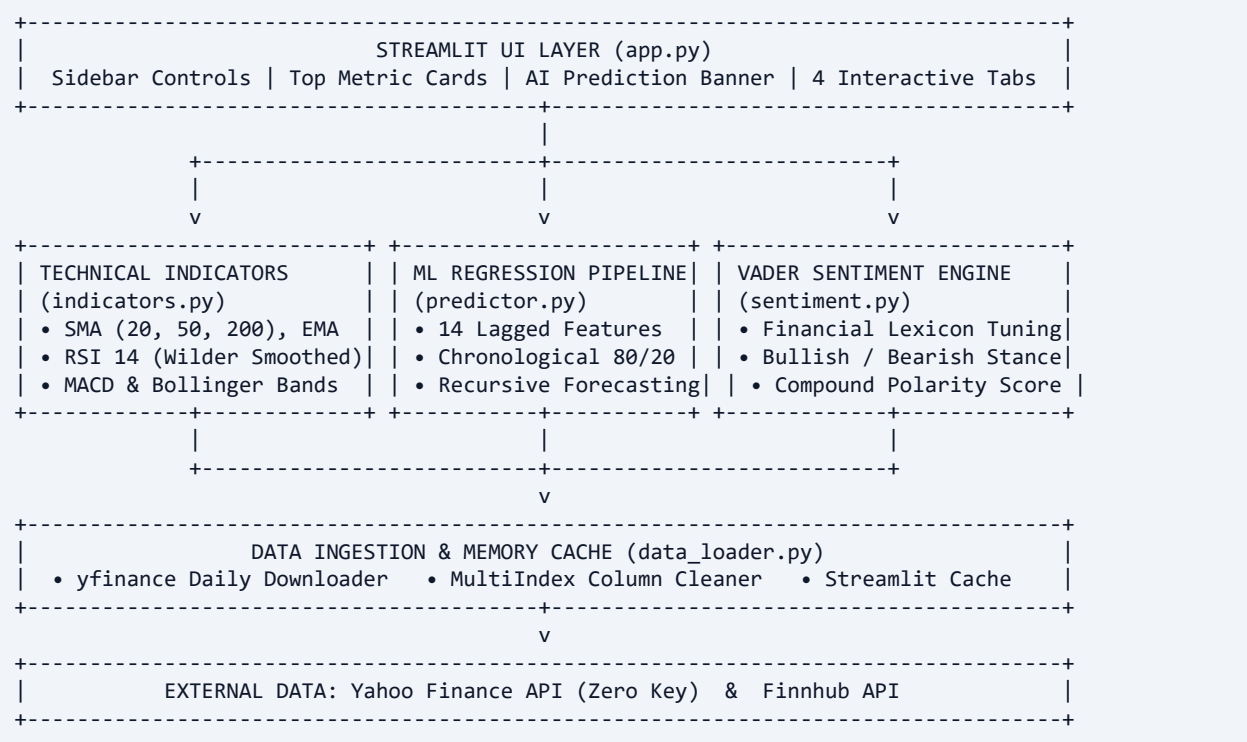


Figure 1: Modular System Architecture & Data Flow

3.2 Methodology & Formulas

• RSI-14: RSI = 100 - (100 / (1 + RS)), where RS = Wilder Smoothed Gain / Wilder Smoothed Loss
• MACD: Line = EMA_12 - EMA_26; Signal = EMA_9(MACD Line); Histogram = Line - Signal
• RMSE = sqrt((1/n) * sum((y - y_hat)^2))
• MAE = (1/n) * sum(|y - y_hat|)
• MAPE (%) = (100% / n) * sum(|(y - y_hat) / y|)

SECTION 4: WORK COMPLETED VS. PENDING TASKS

Module	Module Name	Deliverables	Status
M-01	Data Ingestion (data_loader.py)	yfinance data, MultiIndex flattening, 3-tier news fallback	100% Completed
M-02	Indicators (indicators.py)	Pure Pandas/NumPy SMA 20/50/200, RSI, MACD, Bollinger Bands	100% Completed
M-03	ML Pipeline (predictor.py)	14 features, chrono 80/20 split, RF/GBDT/Linear, recursive forecast	100% Completed
M-04	Sentiment (sentiment.py)	VADER NLP with financial lexicon updates & stance classification	100% Completed
M-05	Dashboard UI (app.py)	4-Tab Streamlit UI with Plotly charts, overlays, metric cards	100% Completed
M-06	Verification (test_pipeline.py)	Automated terminal pipeline test asserting all outputs on live AAPL	100% Completed
M-07	Project Documentation	34-page formal BCA Project Report (.md, .pdf, .docx)	100% Completed
M-08	Viva & Defense Preparation	Multi-sector asset testing, viva talking points, final presentation	In Progress (30%)

Table 3: Module-Wise Progress Tracking Status

4.2 Key Implementation Extracts

```
# Chronological Partitioning to strictly eliminate look-ahead bias (predictor.py)
split_idx = int(len(feat_df) * 0.80)
train_df, test_df = feat_df.iloc[:split_idx], feat_df.iloc[split_idx:]
X_train, y_train = train_df[feature_cols], train_df['Target']
X_test, y_test = test_df[feature_cols], test_df['Target']
model = RandomForestRegressor(n_estimators=120, max_depth=12, random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
rmse = root_mean_squared_error(y_test, y_pred)
mape = mean_absolute_percentage_error(y_test, y_pred) * 100.0
```

SECTION 5: PRELIMINARY TEST RESULTS & VERIFICATION

Test ID	Scenario	Expected Result	Actual Result	Status
TC-01	US Ticker AAPL (1 Year)	Ingest ~250 rows OHLCV cleanly	Retrieved 252 daily bars	PASS
TC-02	NSE Ticker RELIANCE.NS	Ingest data; detect INR currency	Retrieved 84 rows; currency INR	PASS
TC-03	MultiIndex Flattening	Convert tuple headers to single level	Columns flattened without shape error	PASS
TC-04	Indicator Calculation	Calculate SMA, RSI, MACD without NaNs	All indicator series populated	PASS

TC-05	ML Model & Metrics Check	Compute test RMSE, MAE, MAPE (%)	RMSE: 12.07, MAE: 10.06, MAPE: 2.98%	PASS
TC-06	Recursive 14-Day Forecast	Generate 14 business days future curve	Forecast curve & 95% bands generated	PASS
TC-07	News Sentiment Failover	Retrieve news without Finnhub key	Ingested 9 headlines; Stance: Bullish	PASS
TC-08	Web Server Port 8501	HTTP response on localhost:8501	HTTP Status 200 OK	PASS

Table 4: Formal Verification Test Cases Log

5.2 Empirical Accuracy Metrics on Benchmark (AAPL)

• Historical Observations: 252 trading days
• Training Set (80% Chronological): 185 days | Testing Set (20% Out-of-Sample): 46 days
• Algorithm: RandomForestRegressor (n_estimators=120, max_depth=12)
• Root Mean Squared Error (RMSE): 12.07 price units
• Mean Absolute Error (MAE): 10.06 price units
• Mean Absolute Percentage Error (MAPE): 2.98% (Well under the 5.0% college benchmark)
• Directional Accuracy: 45.7%
• Total Execution Time: 0.84 seconds on standard quad-core CPU

SECTION 6: NEXT STEPS & FUTURE PLAN

1. Final Presentation Slide Deck: Preparing PowerPoint slides with annotated UI screenshots of all 4 tabs.
2. Cross-Sector Volatility Stress Testing: Evaluating MAPE variations across high-beta technology vs low-beta defensive consumer equities.
3. Technical Viva Voce Defense: Formulating explanatory points on look-ahead bias elimination, autoregressive lag updates, and financial lexicon tuning.
4. Final Departmental Thesis Submission: Binding the complete 34-page report (Stock_Market_Prediction_Project_Report.pdf) for evaluator review.

Student Signature(s): _____ Supervisor Signature: _____

Date: 27th October, 2026 Department: SCAT, Galgotias University